

Assignment 5

This exercise sheet is devoted to the implementation and analysis of root finding and fixed point methods for nonlinear scalar equations.

Graded exercise

Exercise 19 (Fixed point iteration)

We want to find numerically the solution x to

$$x = (x + 1)e^{-x}, \quad x \geq 0. \quad (1)$$

To do this, we use the fixed point iteration $x_{k+1} = g(x_k)$, $k \geq 0$.

- (a) Implement a PYTHON function `fixed_point` that implements the fixed point iteration. As stopping criterion, use a tolerance on the absolute error between the last two iterates *and* a maximum on the number of iterations N_{iter} . The PYTHON function should accept, in input, the function g of which we are looking for the fixed point, an initial guess x_0 for the fixed point, the tolerance for the absolute error and the maximum number of iterations allowed (the last two inputs are needed for the stopping criterion). The PYTHON function should output two quantities: the final approximation to the fixed point and an array with all iterates x_k , for $k = 0, \dots, N_{iter}$. Note that, although the first output is the last entry of all iterates (second output), it might be easier for an user of the code if we return the two outputs separately.
- (b) Write a PYTHON script that calls `fixed_point` to solve (1) and prints on screen both the approximate fixed point and the value of g there. If your implementation is correct, the two values should (almost) coincide. When calling the fixed point routine, you can use a tolerance of 10^{-9} , a zero initial guess and a maximum of 1000 iterations.

Hint: To define the given g in PYTHON, you can use a `lambda` function.

- (c) Add some lines of code to the script from (b) to compute $\frac{|x_{k+1}-x_k|}{|x_k-x_{k-1}|}$ for $k \geq 1$. Additionally, print the obtained values on screen. For k sufficiently large, the obtained values approximate the rate L in the fixed point convergence.

- (d) Motivate the observed values in the previous task with what we know about theoretical properties of the fixed point iteration.

Hint: For a better motivation, it helps to add a few lines of code to the script to evaluate g' at one or few selected locations. Solving task (a) in Exercise 20 can provide intuition for how to choose these locations.

Other exercises

Exercise 20 (Aitken's acceleration)

In this exercise, we look at a method to accelerate the convergence of a linearly convergent fixed point iteration.

- (a) Let $g : [a, b] \rightarrow I$, with $I \subseteq [a, b]$, be such that $g \in C^1([a, b])$ and $\max_{x \in [a, b]} |g'(x)| := K < 1$. Denote by $(x_k)_{k \geq 0}$ the sequence defined by the fixed point iteration $x_{k+1} = g(x_k)$, $k \geq 0$, for a given initial guess $x_0 \in [a, b]$. Use the fixed point theorem to show that

$$\lim_{k \rightarrow \infty} \frac{x_{k+1} - x_*}{x_k - x_*} = g'(x_*), \quad (2)$$

where $x_* \in [a, b]$ is such that $g(x_*) = x_*$. The quantity $|g'(x_*)|$ is called *asymptotic convergence factor*.

Hint: Use the definition of the iterates and the mean value theorem.

- (b) We denote by λ an approximation of $g'(x_*)$ to be suitably determined. Using (2), we have, for $k \geq 1$,

$$\begin{aligned} x_* &\approx \frac{x_k - \lambda x_{k-1}}{1 - \lambda} = \frac{x_k - \lambda x_k + \lambda x_k - \lambda x_{k-1}}{1 - \lambda} \\ &= x_k + \frac{\lambda}{1 - \lambda} (x_k - x_{k-1}). \end{aligned} \quad (3)$$

Aitken's method provides a simple way to approximate λ so as to accelerate the convergence of the sequence $(x_k)_{k \geq 0}$ to x_* . For this, we consider, for $k \geq 2$, the ratio

$$\lambda_k := \frac{x_k - x_{k-1}}{x_{k-1} - x_{k-2}}. \quad (4)$$

Show that

$$\lim_{k \rightarrow \infty} \lambda_k = g'(x_*).$$

Hint: Note that, for k sufficiently large, $x_{k+2} - x_* \simeq g'(x_*)(x_{k+1} - x_*)$.

- (c) Using, in each iteration, (4) instead of λ in (3), we can define the new sequence $(\hat{x}_k)_{k \geq 2}$ given by

$$\hat{x}_k = x_k - \frac{(x_k - x_{k-1})^2}{(x_k - x_{k-1}) - (x_{k-1} - x_{k-2})}, \quad k \geq 2. \quad (5)$$

This relation is known as *Aitken's extrapolation formula*. Defining

$$\Delta x_k = x_k - x_{k-1}, \quad \Delta^2 x_k = \Delta(\Delta x_k) = \Delta x_k - \Delta x_{k-1},$$

we can rewrite (5) as

$$\hat{x}_k = x_k - \frac{(\Delta x_k)^2}{\Delta^2 x_k}, \quad k \geq 2, \quad (6)$$

which explains why (5) is commonly known as *Aitken's Δ^2 method*.

For the convergence of Aitken's method, it is useful to rewrite it as a fixed point iteration with the function

$$g_\Delta(x) = \frac{xg(g(x)) - (g(x))^2}{g(g(x)) - 2g(x) + x}, \quad (7)$$

where g is as in (a). Show that, if g is differentiable in x_* and $g'(x_*) \neq 1$, then

$$\lim_{x \rightarrow x_*} g_\Delta(x) = x_*.$$

Moreover, it can be shown that the fixed points of (7) coincide with those of g even in the case where x_* is a multiple root of $f(x) = g(x) - x$. It can also be shown that, for linearly convergent fixed point iterations, Aitken's method has quadratic order of convergence.